Progress Report: Resolution of Ambiguous Peptides and Protein Groups (Next Steps #1)

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Project: EV Uptake Proteomics (PC3 vs. PC3M)
Date: November 12, 2026

Overview

This report addresses Task #1, focusing on the systematic resolution of ambiguous, multi-mapping peptide identifications. Rather than discarding shared peptides (which causes substantial quantitative signal loss) or arbitrarily assigning them to a single accession (which introduces false biological certainty), a deterministic, parsimony-based curation pipeline is implemented. This approach rescues justifiable peptide signals using strict sequence evidence and groups unresolved paralogs into standardized multigene families.

Empirical Ground Truth (Step 7.1)

As a baseline from the initial data processing, uniquely mapping peptides were separated from shared entries. Because a peptide matching only a single protein entry provides direct experimental proof of that protein's physical presence in the extracellular vesicles, these entries served as our internal benchmark. By filtering the cleaned data by mapping count, we confirmed 372 genes and set aside 73 unclear peptide records to review separately.

Reference Database Re-mapping (Step 7.2)

Because each row in the original file listed only one Master Accession, it missed key details about alternative protein matches and hid potential paralogs. To find all possible biological sources without bias, every ambiguous peptide sequence was matched against the full, reviewed human Swiss-Prot proteome from UniProtKB. Searching all known human proteins this way provided us with a complete list of candidate genes and accession numbers for each peptide.

Parsimonious Classification and Variant Merging (Step 7.3)

Comparing the candidate genes against the 372 confirmed genes led to three outcomes:

1. **5 peptides** matched several database entries that all belong to the same gene (splice variants), so we kept them under that single gene.
2. **40 peptides** matched multiple genes in theory, but only one candidate was actually confirmed in our sample by unique peptides. Using the simplest explanation (Occam’s razor), these peptides were assigned to the only confirmed protein.
3. **28 peptides** matched multiple confirmed genes at once or lacked unique peptide evidence, so we could not assign them to a specific gene without guessing.

At the same time, two technical modification variants of the same peptide sequence ("SEVDMLK" in *ANXA2*, unmodified and methionine-oxidized) were consolidated by summing their reporter intensities and spectral counts. Consolidating this duplicate reduced the 45 rescued records by one (45 - 1 = 44).

Standardized Multigene Family Grouping (Step 7.4)

Rather than making assumptions about tissue expression or biological plausibility, an agnostic approach was adopted for the remaining 28 ambiguous peptides. Since assigning these peptides to a single gene would require unsupported claims, their quantitative data was preserved by grouping peptides with identical candidate sets into multigene families (e.g., *Family_CLDN_4* or *Family_CAV_2*).

A dedicated reference table maps each family back to its constituent genes. While these families cannot serve as definitive single-gene biomarkers on their own, they retain valuable signal that can support orthogonal evidence (or be resolved later if orthogonal data confirms that only one candidate protein is present in the studied cell type). In total, this organized the 28 unresolved peptides into 19 distinct multigene families:

Family Name	Gene Count	Gene Composition						
Family_ACTB_2	2	ACTB	ACTG1					
Family_CAV_2	2	CAV1	CAV3					
Family_DDX_2	2	DDX3X	DDX3Y					
Family_EEF_2	2	EEF1A1	EEF1A1P5					
Family_PGAM_2	2	PGAM1	PGAM4					
Family_PRPH_2	2	PRPH	VIM					
Family_PSMA_2	2	PSMA7	PSMA8					
Family_RPLP_2	2	RPLP0	RPLP0P6					
Family_RPS_2	2	RPS27	RPS27L					
Family_TUBA_2	2	TUBA1B	TUBA4A					
Family_ATP_3	3	ATP5MC1	ATP5MC2	ATP5MC3				
Family_EIF_3	3	EIF5A	EIF5A2	EIF5AL1				
Family_H2AC_3	3	H2AC25	H2AC4	H2AC6				
Family_H2BC_3	3	H2BC1	H2BC13	H2BC26				
Family_ACTA_4	4	ACTA1	ACTA2	ACTC1	ACTG2			
Family_CLDN_4	4	CLDN3	CLDN4	CLDN6	CLDN9			
Family_HNRN_5	5	HNRNPC	HNRNPCL1	HNRNPCL2	HNRNPCL3	HNRNPCL4		
Family_H2AC_7	7	H2AC11	H2AC12	H2AC14	H2AC18	H2AC20	H2AC7	H2AJ
Family_H2BC_7	7	H2BC11	H2BC17	H2BC19P	H2BC20P	H2BC21	H2BC26	H2BC3

Quality Control and Orthogonality Verification (Step 7.5)

The 44 rescued unique peptides and the 28 unresolved family peptides were combined into a final curated set of 72 records (44 + 28 = 72).

Before calculating gene-level abundance, two automated quality checks were performed: verifying sequence uniqueness within the table and checking for overlap with our primary unique peptides. Both checks passed with zero duplicates and zero overlap. This confirms that quantitative signal is preserved across 20 rescued genes and 19 multigene families without any risk of double counting during downstream roll-up.